

Design and Development of a Faculty Performance Documentation and Evaluation System to Support Evidence Based Decision –Making



A Capstone Project
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CHAPTER 1

INTRODUCTION

The Background of the Study

In the global landscape of higher education, deans are playing an important role in achieving the quality of education, development of faculty and academic excellence. However these deans nowadays are facing a lot of challenges in managing their faculty performance amid of rising expectations regarding accountability, transparency, and data-based decision making. Nkata, J. L., & Ssempebwa, J. (2024), Makerere University, Uganda, highlighted the strenuous nature of deans' roles, emphasizing the difficulty in balancing faculty and administrative demands. Deans often find themselves caught between the expectations of faculty and the directives of the central administration, leading to role conflict and ambiguity. Hussain et al. (2025), said that to be fair, consistent, and successful, performance review systems need to be both quantitative and qualitative.

In the case of Philippine Higher education, these difficulties are worsen because of the limited digital resources, administrative inefficiencies, and inconsistent standards in faculty evaluation practices. The Second Congressional Commission on Education (EDCOM 2) highlighted in its 2025 report that many State Universities and Colleges (SUCs) continue to rely on the outdated and fragmented systems, which lead to weak governance and poor data utilization. The Philippine Institute for Development Studies (PIDS) further observed that despite the increasing enrollees in the higher education, the quality assurance and institutional management is still inconsistent, with some SUC are still struggling with the implementation of strong and effective systems for evaluation success.

Colleges and universities outside of major cities face additional challenges, such as perceived biases, a lack of transparency in the evaluation process, and delays in faculty performance feedback (Abejo, 2024). Many of these schools lack clear guidelines for research and evaluation, provide teachers with excessive work outside of class, and have even fewer resources, which demotivates teachers and hinders the success of the school (Dela Cruz & Dela Cruz, 2024). Additionally, a large number of regional institutions still conduct faculty reviews using manual, paper-based processes, which exacerbates inefficiencies and delays crucial administrative decisions (Loyola, 2023). Therefore, regional higher education institutions face even more serious issues like a lack of resources and unclear procedures in addition to the same issues facing the rest of the nation, such as inconsistent standards and weak leadership (Abejo, 2024; Dela Cruz & Dela Cruz, 2024; Loyola, 2023).

At the institutional level, Dr. Emilio B. Espinosa Sr. Memorial State College of Agriculture and Technology (DEBESMSCAT) is also facing this kind of challenges. One of the most urgent issues is the manual processing and reconciliation of faculty performance data, which is time-consuming, prone to error, and lacks transparency. This method contributes to delays in evaluation, inconsistencies in documentation, and difficulties in aligning faculty outputs with institutional priorities. As a result, not only are deans burdened with administrative workload, but overall academic planning, resource allocation, and institutional reporting are also compromised.

The Individual Performance Commitment and Review (IPCR), is a standard tool required by the Civil Service Commission under the Strategic Performance Management System (SPMS). It is an essential component of faculty performance evaluation in every SUC. Part of the IPCR process is setting performance goals, keeping track of accomplishments, and evaluating results based on Office Performance Commitment and Review (OPCR) goals. But there are big problems in preparing IPCRs manually, such as duplication of data, lack of real time monitoring, and having trouble in compiling all reports for submission. These issues make it harder to keep track of success and make decisions.

These challenges are not exclusive to DEBESMSCAT. Other Philippine institutions report similar issues. For instance, a study at Pangasinan State University found that faculty members with multiple roles struggled with preparing IPCRs due to overlapping responsibilities and the absence of standardized performance computation models. The study proposed a mathematical model, evaluated using the Technology Acceptance Model, which significantly improved user satisfaction (Cenas, 2022). Similarly, at Leyte Normal University, researchers observed that although faculty performance ratings were generally satisfactory, the process heavily depended on manual inputs, which affected the reliability and consistency of results (Caluza et al., 2017).

The IPCR workflow typically begins with setting individual targets aligned with the institution's OPCR. Faculty members then document their accomplishments throughout the performance period, which are validated by their respective deans and submitted to the Performance Management Team. Key performance areas—namely instruction, research, extension, and production—carry varying weights, with instruction often taking precedence due to its direct link to student outcomes and accreditation requirements.

Given the systemic and institutional inefficiencies, there is a clear and urgent need for a digital solution that streamlines faculty performance documentation and evaluation, reduces administrative burden, and enhances data accuracy and usability. In response, this capstone project proposes the design and development of a Faculty Performance Documentation and Evaluation System tailored to DEBESMSCAT's operational context, while aligning with national standards such as SPMS and OPCR/IPCR protocols. The system aims to support evidence-based academic planning, foster transparency, and empower academic leaders in making informed decisions.

Critical Success Factors

- 1. User-Centered System Design** - The system is user-friendly and tailored to the faculty workflows. Early user's engagement during the design phase ensures the system success.
- 2. Accurate and Secure Data Management** – The system ensures data integrity, confidentiality, and security when handling performance data. It includes validation mechanism and role-based access controls.
- 3. IPCR and SPMS Standards Alignment** – The system complies with the Civil Service Commission's guidelines for IPCR documentation. It supports the standard templates, rating scales and performance indicators.

4. **Automation of Manual Process** – Automating data entry and report generation will reduce errors and faculty burden. This includes the consolidation of monthly and quarterly report, and auto calculation of overall rating.
5. **System Acceptability and Adoption** – This system is accepted by the intended users which is measured through usability testing, feedback surveys, and pilot implementation. Training and support materials was provided in facilitating the adoption.
6. **Scalability and Integration** – The system is dynamic to accommodate changes or possible expansion to other colleges.
7. **Intellectual Property (IP) Feasibility** – The system uses original design and functions to qualify for IP protection. Documentation, source code management and innovation is maintained for IP filing.
8. **Data Driven Reporting Capability** – The system is generating analytics and reports that support decision making by Deans. Key features like dashboards, and performance summaries are very useful.
9. **Compliance with Institutional Policies** – The system is aligned with DEBESMSCAT quality policy and support institutional goals.

Problem/Opportunity Definition

At DEBESMSCAT, faculty members and deans face difficulties with regards to performance data reconciliation, IPCR documents consolidation, and aligning outputs with institutional goals. The absence of a centralized digital system results in redundant documentation, limited access to historical data, and challenges in consolidating reports for OPCR submissions. This limits the institution in making more informed decisions, allocating resources and strategic planning.

Despite the digital technologies availability, currently DEBESMSCAT has no system that would address this needs of faculty performance documentation and evaluation. Also, the lack of innovations like this limits the potential for intellectual Property production, which could contribute to institutional goals.

Specifically, it seeks to answer the following questions:

1. What are the limitations of the current manual system used for faculty performance documentation and IPCR preparation at DEBESMSCAT?
2. How can the proposed system improve the efficiency, accuracy, and acceptability of faculty performance evaluation among stakeholders?
3. What innovative features can be integrated into the system to qualify it for intellectual property protection?

General Objective:

The general objective of this capstone project is to design and develop a Faculty Performance Documentation and Evaluation System for DEBESMSCAT that facilitates evidence-based decision-making by automating the collection, analysis, and reporting of faculty performance data. Specifically, it aims to:

1. To develop a functional web-based system that enables efficient documentation, monitoring, and evaluation of faculty performance based on predefined institutional criteria and metrics.
2. To assess the usability, functionality, and overall acceptability of the developed system among faculty members through user testing and feedback mechanisms.
3. To create a patent draft document that outlines the unique features and technical specifications of the developed system.

Significance of the Study

This study is significant as it addresses a critical gap in the performance evaluation process of faculty members within State Universities and Colleges (SUCs), particularly at DEBESMSCAT. By designing and developing the system tailored to the documentation and evaluation of faculty performance, the study contributes to the modernization of institutional processes and supports the broader goal of evidence-based decision-making in higher education.

1. **For the Institution (DEBESMSCAT):** The system will help the institution's efficiency by automating the preparation and consolidation of IPCR reports. This will reduce the burden on deans and faculty, reduce errors, and improve the timeliness and accuracy of performance evaluations.
2. **To the Faculty Members:** All Faculty will benefit from an efficient and transparent evaluation process. The system will allow them to monitor their performance in real-time, align their outputs with institutional goals, and receive timely feedback.
3. **To the Deans:** Deans will gain access to reliable performance data that can inform decisions related to promotions, workload distribution, and faculty development programs.
4. **To the Higher Education Sector:** The study contributes to the ongoing digital transformation of SUCs in the Philippines. It aligns with national efforts to improve governance, transparency, and quality assurance in higher education, as emphasized by EDCOM 2 and the Civil Service Commission.
5. **To Future Researchers and Developers:** This project serves as a foundation for future innovations in academic performance management systems. It opens opportunities for further research in areas such as predictive analytics, AI-assisted evaluation, and integration with learning management systems.
6. **To Intellectual Property Development:** By incorporating original design and functionality, the system has the potential to be registered as intellectual property. This not only adds value to the institution but also promotes innovation and recognition in the field of educational technology.

The scope of the Study

This study focuses on the design and development of a Faculty Performance Documentation and Evaluation System specifically tailored for Dr. Emilio B. Espinosa Sr. Memorial State College of Agriculture and Technology (DEBESMSCAT). The system aims to automate and streamline the preparation, submission, and evaluation of

the Individual Performance Commitment and Review (IPCR) and Office Performance Commitment and Review (OPCR) forms.

The system will include the following core features:

- Digital encoding and submission of IPCR data
- Manual input of ratings per performance criterion by authorized evaluators
- Automated computation of the overall performance rating based on weighted criteria
- Real-time tracking of faculty accomplishments
- Role-based access for faculty, deans, and staff.
- Report generation for performance summaries and OPCR consolidation
- Basic analytics to support evidence-based decision-making

The system will be deployed at the College of Arts and Sciences, DEBESMSCAT and will primarily serve Faculty and Dean.

Delimitations of the Study

- The system is developed and tested only within DEBESMSCAT, and may not immediately reflect the needs or workflows of other SUCs or private institutions.
- While individual criterion ratings are manually inputted, the system automatically calculates the overall performance rating to ensure consistency and reduce human error.
- The evaluation will focus on instructional, research, extension, and production functions as outlined in the IPCR, but will not include other HR-related modules such as payroll, leave management, or promotion tracking.

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

Related Literature

The growing emphasis on evidence-based decision-making in higher education has transformed the way institutions evaluate faculty performance. The literature reveals how academic organizations worldwide are adapting to new methods of documentation, automation, and holistic assessment. This synthesis categorizes relevant research into four major themes, aligned with the goals of your proposed Faculty Performance Documentation and Evaluation System.

Data-Driven and Contextualized Evaluation Models

Globally, there is a significant shift toward data-driven and institution-specific faculty evaluation systems. Hussain, Lee, and Patel (2025) highlighted the role of analytics-powered platforms in delivering personalized assessments that balance transparency and academic rigor. Wang (2023) illustrated how cultural and institutional contexts influence evaluation approaches in elite institutions such as Harvard and Peking University. Santiago and Benavides (2009) and Isoré (2009) both cautioned against rigid, standardized models, instead recommending adaptive frameworks suited to individual institutions. Khan (2013) advocated for developmental evaluations that foster faculty growth rather than impose punitive measures. Complementing these conceptual insights, Benosa and Oñate (2023) and Dela Fuente and Dela Fuente (2022) introduced analytics-integrated platforms that empower institutional decision-making. Meanwhile, Rapacke (2024), Thompson (2025), and Tinch (2025) emphasized the need for originality and innovation in system design to meet the technical requirements for intellectual property eligibility.

In the Philippine context, studies similarly call for automated and analytics-enhanced evaluation systems adapted to local needs. Bueno (2023) reported inconsistent faculty evaluations at Columban College due to the lack of standard tools. Cenas (2022) responded to the complexity of multiple faculty roles by proposing a mathematical model for IPCR accuracy validated using the Technology Acceptance Model. Dela Fuente and Dela Fuente (2022) demonstrated how analytics integration improved assessment reliability and minimized human error. At ISPSC, Cinizan and Valdez (2023) enhanced their Faculty Performance Evaluation System by embedding analytics features, achieving a very high usability rating. Romblon State University implemented intelligent evaluation mechanisms such as Common Criteria Evaluation (CCE) and Qualitative Contribution Evaluation (QCE), supported by analytics to streamline assessments. The Polytechnic University of the Philippines linked online evaluation results to teaching load and faculty promotions, reinforcing performance-based advancement. National frameworks such as CHED's Institutional Sustainability Assessment (ISA) toolkit and Self-Evaluation Document (SED), along with DepEd's IPCRF system and the Civil Service Commission's Strategic Performance Management System (SPMS), collectively reflect a growing trend toward objective, data-backed, and context-aware faculty evaluations in the country.

Holistic and Multifaceted Faculty Evaluation

Foreign scholars increasingly advocate for comprehensive faculty evaluation systems that consider not just instructional effectiveness but also research productivity,

service engagement, and professional development. Kezar (2018) argued for aligning evaluations with institutional missions and faculty roles, while Darling-Hammond (2013) promoted formative evaluations as tools for professional growth. Weaver and Carlisle (2023) offered implementation strategies for such holistic models, and Kumar (2025) discussed India's UGC framework that ensures equal weight across instruction, research, and service. Brown (2015) and Golde and Dore (2001) highlighted the disconnect between expectations and faculty preparation, proposing feedback systems to bridge the gap. Additionally, Lohman (2021), Bernstein (2008), and Boyer (1996) supported integrating mentoring, peer review, and scholarly identity into evaluation structures to ensure meaningful, development-oriented assessments.

Locally, multiple studies reflect this call for more holistic and multidimensional faculty evaluations. Caluza et al. (2017) found that the reliability of faculty assessment at Leyte Normal University was compromised by manual input processes, advocating for digital systems that integrate multiple indicators. Patimo (2020) observed that while SUCs in Eastern Visayas employed rating scales, they often lacked automation and qualitative assessment tools. EDCOM 2 (2024) highlighted faculty evaluation as essential for promoting both institutional growth and individual professional development. CHED's ISA framework positioned faculty evaluation as a cornerstone of institutional sustainability, focusing on quality teaching, governance, and community involvement. DepEd's Results-Based Performance Management System (RPMS) and IPCRF modules emphasized the importance of development planning and formative feedback. The CSC's SPMS Guidebook introduced a four-stage cycle—planning, monitoring, review, and rewarding—designed to support continuous improvement. Likewise, evaluation manuals from ASSCAT and PCSO link performance assessments to staff incentives and career development, while CHED Region 2 provided templates that help institutions align evaluations with broader academic goals.

Usability and System Innovation

The usability of faculty evaluation systems is a critical factor in their success, and international research has emphasized the importance of user-centered design (UCD) principles. Cruz et al. (2021) demonstrated that UCD significantly enhances user adoption and long-term engagement with performance platforms, a conclusion further supported by Alao et al. (2022) through practical application in university systems. Yesa, Putri, and Noer (2024), Sani et al. (2023), and Orallo (2025) all highlighted how intuitive, design-thinking-based interfaces on academic websites and mobile applications improve efficiency and accessibility, ultimately encouraging more meaningful participation from users.

In the local setting, several Philippine institutions have demonstrated how system innovation and usability improvements lead to better performance monitoring. Tarlac State University's Automated Faculty Evaluation System improved transparency and reduced administrative workload through an intuitive design. At ISPSC, the enhanced FPES utilized Agile development and data analytics to achieve high usability scores. Central Mindanao University's system automated faculty performance monitoring and supported multi-source feedback mechanisms. NVSU developed a web and mobile-based platform that enabled students to evaluate faculty and access real-time performance dashboards. Salas (2015) built a mobile-responsive evaluation system that received strong ratings for functionality and portability. Plata and Facun (2020) implemented a paperless feedback

platform that simplified the student evaluation process. Ramos et al. (2020) designed a LAN-based appraisal system that improved documentation efficiency and reduced evaluation time. Celis and Pagatpatan (2019) applied the Analytic Hierarchy Process (AHP) algorithm to increase the accuracy of evaluation decision-making, while Fadera and Enriquez (2018) integrated intelligent analytics into their systems to accelerate data processing and reduce errors.

IP Eligibility and System Originality

The global literature underscores the importance of originality and technical innovation in creating systems that are eligible for intellectual property (IP) protection. Rapacke (2024) explained that systems must solve identifiable technical problems and incorporate method-based uniqueness to qualify for protection. Thompson (2025) discussed how the Alice Two-Part Test determines whether a system's features go beyond abstract ideas, which is essential for patentability. Tinch (2025) emphasized innovations such as improved interactivity and novel analytics as enhancements that support IP claims. Collins (2024) and InspireIP (2025) both reinforced that comprehensive documentation and strategic technological advancements are fundamental to obtaining software IP rights. ISO (2025) highlighted the role of security standards and originality criteria in safeguarding digital systems from replication and unauthorized use.

In the Philippines, recent developments also reflect efforts to build evaluation systems that meet the standards of IP eligibility. Dela Fuente and Dela Fuente (2022) highlighted that their analytics-based system was designed with original features tailored to institutional faculty evaluation, making it a candidate for IP registration. Cinizan and Valdez (2023) documented the unique development process of their FPES, which included modular architecture and advanced data visualization components. Romblon State University integrated custom algorithms and analytics dashboards to strengthen the system's claim to technical originality. CHED's ISA toolkit further encourages higher education institutions to innovate and document their internal quality assurance mechanisms—including faculty evaluation platforms—as a step toward institutional development and intellectual property recognition.

Related Systems

Automated and Analytics-Driven Systems

Globally, institutions are embracing systems that automate faculty evaluation while integrating powerful analytics. Tools like Cflow Faculty Evaluation Platform and EduRate utilize AI features—such as OCR, decision trees, and data visualization—to generate performance insights. Multi-criteria decision-making models like AHP, TOPSIS, and Grey MARCOS enhance objectivity across complex faculty roles. These systems exemplify how automation and data science can refine evaluation practices and support evidence-based administration.

In the Philippine context, SUCs are rapidly adopting similar tools. ISPSC's enhanced Faculty Performance Evaluation System uses analytics and Agile methods to streamline reviews and received high usability marks. Romblon State University's intelligent evaluation model applies Common Criteria and Qualitative Contribution Evaluation to quantify faculty output. DepEd's IPCRF templates and CHED's ISA framework emphasize digital tracking and strategic alignment. These local systems mirror

global trends while addressing region-specific challenges like multi-role faculty and resource constraints.

Usability and Interface-Focused Systems

International studies highlight how user-centered design principles boost system adoption in academic settings. Platforms such as Visual Analytics for Teaching Practice and dashboards embedded in EduRate rely on intuitive navigation, real-time feedback, and mobile responsiveness to encourage engagement. AHP-based decision support and clean design layouts help stakeholders provide input easily and interpret results quickly.

Filipino institutions are equally focused on interface design that caters to users. Tarlac State University's AFES features student-facing tools, checkbox evaluations, and streamlined reporting. Central Mindanao University and NVSU use mobile-accessible systems with dashboards tailored for faculty, admins, and students. Systems developed by Ramos, Plata, Salas, and Celis prioritized intuitive design, portability, and data reliability. The emphasis on ease-of-use aligns Philippine innovations with global best practices, demonstrating that user experience is as vital as analytical power in faculty evaluation.

IP-Eligible and Innovative Systems

Across foreign literature, originality in system architecture is key to achieving intellectual property protection. Researchers emphasize patentability through modular design, unique algorithms, and documented innovation. Platforms like InspireIP and ISO-compliant models ensure legal eligibility by capturing inventive technical solutions and safeguarding information security. Rapacke, Thompson, and Tinch outline the legal frameworks—such as the Alice Two-Part Test—that systems must satisfy to be protected under IP law.

Philippine developers are rising to the challenge of creating novel systems eligible for IP registration. ISPSC and Romblon State University embedded distinct analytics engines, modular structures, and custom evaluation workflows into their platforms. Documentation efforts and Agile-based development processes showcase originality. CHED's ISA guidance encourages institutions to treat internal quality assurance tools as potentially proprietary assets. The innovative use of algorithms like AHP for faculty ranking further bolsters the claim for IP protection. This shows how Philippine systems are not just following global standards—they're carving out their own space in the realm of educational software innovation.

CHAPTER 3

DESIGN AND METHODOLOGY

Research Design

This study adopts a systems development research approach, focusing on the design, implementation, and evaluation of a Design and Development of a Faculty Performance Documentation and Evaluation System to Support Evidence Based Decision –Making. Grounded in the Information Systems Success Model (ISSM) (DeLone & McLean, 2003), the framework ensures that the system effectively enhances data quality, system usability, and academic decision-making effectiveness.

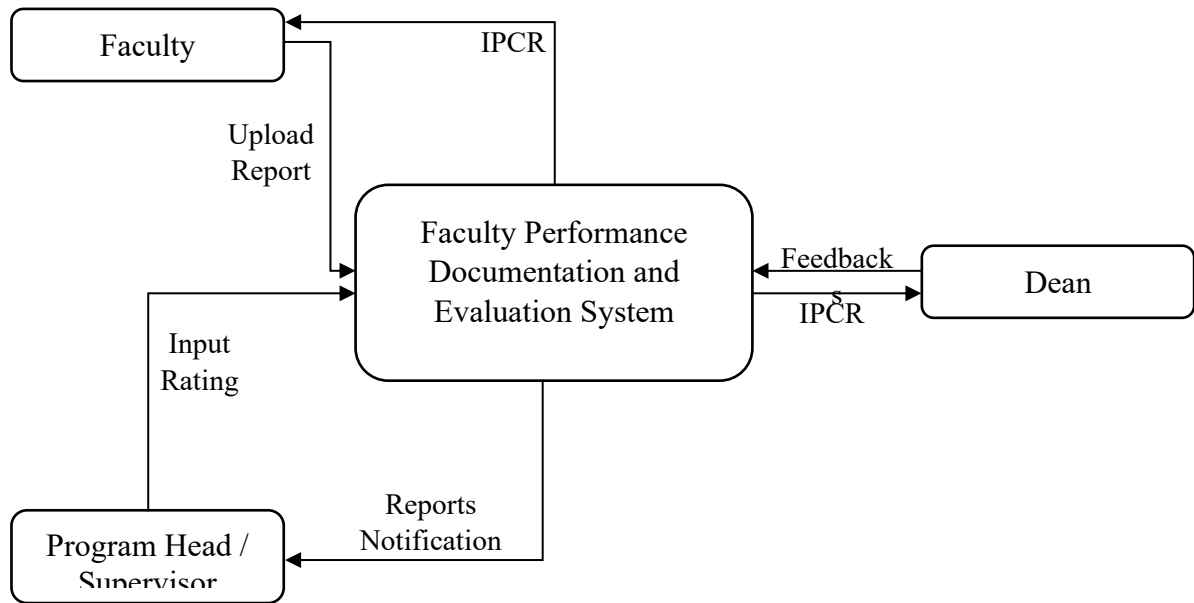


Figure 1. Conceptual Framework

The system has three main entities which is the Faculty, Program Head / Immediate Supervisor and the Dean. Faculty will upload its Core function reports, and other support function reports to the system for the system to consolidate it. The Program Head or Immediate Supervisor will be notified every time their people will submit reports that needs to be check and rate them based on the criteria given. On the other hand, Dean has the view of all the documents being uploaded and can give feedback based on the data on the dashboard. Once the period is complete, the IPCR will be available in both Faculty and Dean as output and DPCR for Program Head or additional Data for DPCR contribution for other Immediate Supervisor.

Analysis and Design

People Analysis: The current Faculty Performance Evaluation process involves multiple stakeholders across academic and administrative tasks. It includes the Faculty members who prepared and submit Individual Performance Commitment and Review

(IPCR) forms, Program Heads and Supervisors who reviews submissions and assign ratings, and Deans who oversee College performance and consolidates evaluation results.

Process Analysis: The existing process flow is manual and paper-based, beginning with the cascading of targets, distribution of IPCR templates, followed by individual preparation of reports, physical submissions, and manual checking of supervisors. The ratings are consolidated using spreadsheets and printed for administrative review. This process is prone to delays, data entry errors, and inconsistent data formatting, especially during evaluation periods. Decision points – such as report validation, rating approval, and document generations are often bottlenecks due to limited coordination and lack of real time visibility. The absence of centralized tracking make it difficult to monitor submissions status, identify missing reports, or generate summaries on time.

A digital system with automated workflows, submission notification, and users dashboards will streamline this process, reduce redundancy, and improve accountability.

Technology Analysis: The College is still relying on MS Excel and email for performance documentation, without centralized database platform. Faculty reports are stored in local drives or printed archives, which results to labor intensive when retrieve. The network infrastructure supports browser-based access and cloud hosting, but lacks dedicated systems for performance management. Existing hardware—faculty desktops and admin workstations—is sufficient for web-based applications, though some upgrades may be required for older devices. The proposed system is build using HTML, PHP, and Javascript which is compatible to the current infrastructure and offer secure multi-user access, role based dashboards, and report generation. The absence of automated rating calculators, submission tracking, and document archiving are all addressed by proposed system.

Feasibility Analysis and Managerial Impact

Technical Feasibility

The proposed system is technically feasible given the institution's existing infrastructure and available expertise. The development stack - HTML, PHP, JavaScript (jQuery/Ajax), MySQL, and Bootstrap is compatible with the university's browser-based environment and can be deployed on standard web servers. Development will be conducted using Visual Studio Code (VS Code), which is lightweight and widely supported. The institution's IT personnel are familiar with these technologies, ensuring adequate support for deployment and maintenance. Faculty and administrative staff possess basic digital literacy, and targeted training will be provided to bridge any gaps. No third-party vendors are required, as the system will be built using open-source tools and maintained internally. Given the alignment with current systems and the availability of technical resources, the project is deemed technically feasible.

Economic Feasibility

The projected cost for this project is around ₱15,000 which is economically feasible. This budget includes training cost, and contingency fund. Expenses for domain hosting will free because it will be covered by the schools current subscription. Expenses

for Licensing and Vendors fee can be avoided by using open-source technologies and internal development. Because the system expedites the workflows, shortens performance documentation, and lower administrative cost, the return on investment (ROI) is anticipated during the first academic cycle. The project is economically feasible given these cost savings and long term advantages.

Operational Feasibility

The system is well align with the current university workflows and organizational structure. It improve the transparency and efficiency by digitalizing the present IPCR process without affecting essential duties. Real-time dashboards, rating calculators, and automatic submissions will be helpful to instructors and assessors. Feedback loops, systematic training programs, and pilot testing all help to promote user adoption. The system embodies the institutional values of evidence-based governance, accountability, and ongoing development. The implementation of change management tactics, such as advocate networks, awareness campaigns, and helpdesk support, has made the system both operationally and culturally feasible.

Legal/Ethical Feasibility

The system adhere to the Data Privacy Act of 2012 (RA 10173), implementing secure login protocols, encrypted data transmission and role-based access controls. It also obtain the user consent at first login, and privacy policies are clearly explained. The rating procedure has ethical measures to make sure that assessments are based only on authorized IPCR indicators. Transparency and contextual reasoning are made possible via feedback mechanisms. To guarantee continued compliance, audits and privacy impact analyses will be carried out on a regular basis. The system is therefore morally and legally sound.

Managerial Impact

The system will improve the managerial strategic decision making. Dean and Program Head will be able to monitor faculty contributions and efficiently distribute resources using automated documentation and performance analytics. With the use of dashboard, reporting process will be more easier, and manual report consolidation will be change to planning and data analysis. By offering organized, auditable records, the technology facilitates institutional accreditation and CHED Institutional Sustainability Assessment compliance. Aligned with the institution's objectives for academic excellence and quality assurance, the system enhances managerial capability overall.

Change Management Component

The Faculty Performance Documentation and Evaluation System successful implementation depends on the coordinated change Management approach that takes the organizational preparedness and individual adaptability. Key stakeholders are faculty members, program heads and Dean. Faculty members, as primary user's may resist at first due to possible increase of accountability and unfamiliar of the platforms. Program heads and deans may need to adjust their monitoring and reporting procedure, even if they expect to gain from much better workflows.

The impact of the system to the administrators or Dean will have better access to gather data for reporting and decision-making. While the faculties will transition from manual to

automated performance tracking. Despite of the advantages, these requires a careful adjustment.

This approach will be guided by the ADKAR model. College briefing and communication strategies will be used for more awareness. By showcasing the system advantages through pilot testing and consultations, stakeholders desire to use the system will increase. Trainings, workshops and digital modules will be utilizing for acquiring more knowledge about the system. Lastly, Real time technical assistance, and peer coaching will help to strengthen their ability.

Interventions will include a centralized support center for assistance and problems, interactive workshops, and department-level pilot programs run by faculty allies. Continual improvements and frequent feedback gathering will guarantee that the system adapts to user requirements. These strategies aim to reduce opposition, increase institutional trust, and establish the system as a fundamental part of academic governance.

Communication Plan

Effective communication played a critical role in the successful development, deployment, and adoption of the Faculty Performance Documentation and Evaluation System. Clear channels and regular updates were established among key stakeholders to ensure transparency, collaboration, and timely dissemination of information throughout the project lifecycle.

Audience: Faculty Members, Program Heads/Supervisors, Deans, Project Development Team.

Message: Key updates and milestones such as system goals, schedules, training sessions, submission deadlines, evaluation procedures, and feedback opportunities, will all be covered in the message. Guidance on system usage, login credentials, troubleshooting techniques, and policy integration pertaining to IPCR/DPCR documents will also be included.

Channels: Information will be disseminated through the following channels:

Email: for formal announcements, login instructions and documentation guides.

College Meetings: for Q&A, feedback collection and interactive briefings.

Shared Drive: for access of training materials and manuals.

Faculty Group Chats: for reminders and quick alert.

Workshops: for hands-on system walkthroughs and role specific training.

Frequency: Weekly updates during pilot and training phases. Twice a week progress report during development and testing. Monthly summary reports to deans during active evaluation cycles.

Responsible Person:

Project Leader/Developer – for technical updates and system documentation

Program Head – for cascading the information to the faculty

Dean's Office – for College level announcement and policy alignment

Schedule and Budget Summary

Project Phases/Milestones:

Phase 1: Project Initiation & Planning – July 28, 2025 – August 8, 2025

- Milestone A: Finalization of project scope, stakeholder alignment, and approval of system architecture
- Phase 2: System Development and Testing – Aug 1, 2025 – October 20, 2025
- Milestone B: Completion of prototype and internal testing
- Phase 3: Pilot Implementation – Oct 21, 2025 – October 22, 2025
- Milestone C: Deployment to selected departments and feedback collection
- Phase 4: Training and Full Rollout – October 23, 2025 – October 23, 2025
- Milestone D: Conduct of faculty workshops and system activation
- Phase 5: Monitoring and evaluation – October 24, 2025 – October 30, 2025
- Milestone E: Post Implementation review and documentation for college reporting.

Gantt Chart

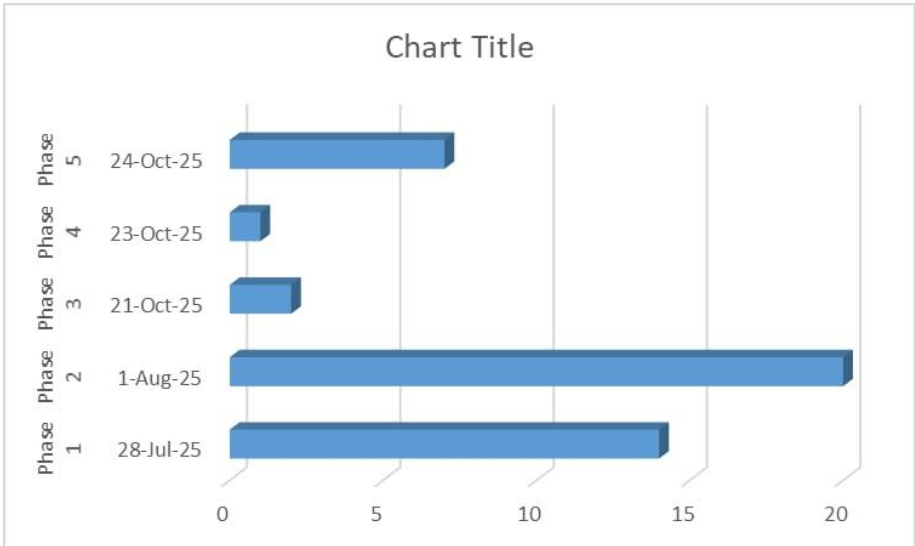


Figure 2. Gantt Chart

Budget

Item	Cost Estimate
Domain & Hosting	₱ 0.00
Dev Tools & Licenses	₱ 0.00
Training	₱ 10,000
Contingency	₱ 5,000
Total	₱ 15,000

Requirement Specification

The successful development and deployment of the Faculty Performance Documentation and Evaluation System required a clear specification of both functional and non-functional requirements. These requirements were gathered during the planning and analysis phases through consultations with institutional stakeholders, benchmarking of existing systems, and alignment with CHED and CSC evaluation frameworks.

Functional Requirements

The system's functional requirements define the essential operations and services it must provide to fulfill its intended role in documenting and evaluating faculty performance:

- **Login and Registration.** Enables authorized users to register and securely log into the system using unique credentials. Role-based access allows faculty, administrators, and evaluators to interact with modules relevant to their responsibilities.
- **Faculty Dashboard and IPCR Tracking.** Displays real-time evaluation status, visual summaries, and detailed breakdowns of submitted data. IPCR forms and scores are tracked across defined evaluation periods.
- **Administrative Control Panel.** Allows administrators to manage user accounts, configure evaluation periods, update criteria weights, and monitor submission activity. Centralized access simplifies institutional oversight.
- **Evaluation Form Submission.** Faculty can document achievements under teaching, research, service, and administrative functions. Forms support file attachments, draft saving, and submission confirmation.
- **Performance Rating Calculator.** Automatically computes evaluation scores based on institutional metrics and assigned weights. Supports multiple rating categories with aggregated totals aligned with SPMS and IPCR standards.
- **Report Export and Print View.** Users can generate performance summaries and IPCR scorecards. Reports are exportable in PDF or printer-friendly formats and accessible per evaluation cycle.

Non-Functional Requirements

Non-functional requirements define the performance standards, security protocols, and system behavior needed to ensure reliability and user satisfaction:

- **System Load Speed.** Web pages and dashboards must load in under two seconds under normal network conditions to support efficient user workflows.
- **Secure Login Protocol.** User authentication is protected through Secure Socket Layer (SSL) encryption and token-based login sessions to prevent unauthorized access and data breaches.
- **Mobile Responsiveness.** The system adapts to various screen sizes and device types, enabling access via desktop computers, tablets, and smartphones without layout distortion.
- **Daily Backup and Restore Features.** System data is automatically backed up each day to cloud storage, ensuring recovery options in case of data corruption or accidental deletion.
- **Concurrent User Support.** The platform supports up to 30 simultaneous users without degradation in performance, ensuring accessibility during peak submission periods.

Project Management Plan

Staffing

To support the successful development, testing, and deployment of the Faculty Performance Documentation and Evaluation System, a multidisciplinary project team was

assembled. Each role was carefully selected based on the technical requirements, documentation demands, and quality assurance standards of the capstone project.

Project Manager. Responsible for overall coordination of project activities, including planning, documentation, and stakeholder communication. Ensure alignment with institutional goals, CHED compliance, and academic standards. He/she must be skilled on strategic planning, stakeholder engagement, project documentation, and leadership.

System Analyst. Conduct requirement gathering, process analysis, and system modeling. Insure functional alignment with faculty evaluation process. Required skills are the following: Business process analysis, data modeling, documentation, and familiarity with IPCR frameworks.

Developer: Builds the system using HTML, PHP, JavaScript (jQuery/Ajax), and MySQL. Implements backend logic, frontend interfaces, and database structures. Ensures system security, performance, and maintainability. He/she must be Proficiency in PHP, MySQL, HTML/CSS, JavaScript, Bootstrap, version control (Git), and secure coding practices.

UI/UX Designer: Responsible for Designing user-friendly interfaces and ensures intuitive navigation across modules. Develops responsive layouts and accessibility features for faculty and administrative users. Required Skills: UI/UX design principles, Bootstrap, wireframing, user testing, and accessibility standards.

Tester: Develops and executes test cases for unit, integration, system, and user acceptance testing. Recognizes bugs, documents issues, and validate fixes before the deployment.

Work Breakdown Structure (WBS)

Code	Task Description	Duration (days)	Dependencies
1.0	Project Initiation	14	-
	1.1 Define Project Scope	5	-
	1.2 Form Team	2	1.1
	1.3 Stakeholder Consultation	5	1,2
2.0	Requirements Analysis	14	1.2
	2.1 Gather Requirements	7	1.2
	2.2 Document Requirements	3	2.1
	2.3 Map Workflows & Validate Specs	4	2.2
3.0	System Design	14	2.2
	3.1 Create Wireframes and UI Mockups	7	2.2
	3.2 Design Database Schema	3	3.1
	3.3 Finalize Architecture and Flowcharts	4	3.2

4.0	Development and Implementation	30	3.3
	4.1 Backend Development (MySQL)	14	3.3
	4.2 Frontend Development (PHP, HTML, Javascript)	7	4.1
	4.3 Integration and Feature Completion	7	4.2
5.0	Testing and Refinement	14	4.3
	5.1 Unit and Integration Testing	7	4.3
	5.2 User Acceptability Walkthroughs	3	5.1
	5.3 Bug Fixes and Refinement	4	5.2
6.0	Deployment	7	5.3
	6.1 Cloud Hosting and Domain Setup	3	5.3
	6.2 System Rollout and Activation	4	6.1
7.0	Documentation and Training	7	6.2
	7.1 Prepare User Manuals and Guides	3	6.2
	7.2 Conduct Training and Orientation	4	7.1
8.0	Project Closure	2	7.2
	8.1 Final Report and Manuscript Submission	1	7.1
	8.2 Post-Deployment Review and Feedback	1	8.1

PERT Diagram

Figure 3. PERT Diagram

Data Modeling

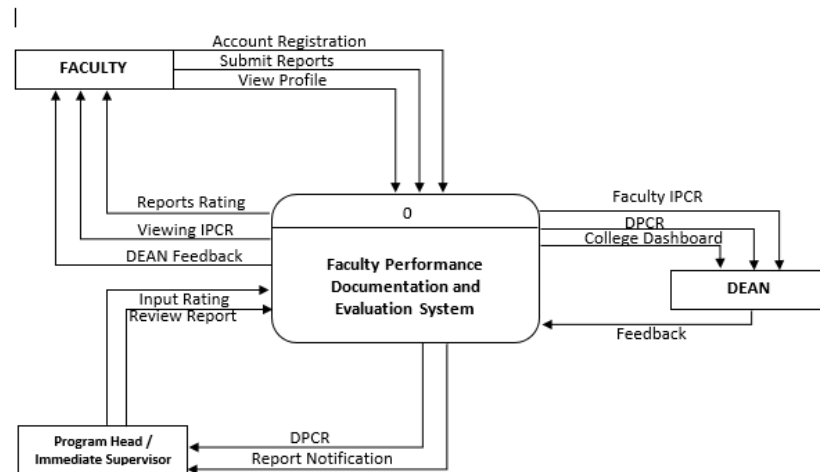


Figure 4. Context Diagram (DFD – Level 0)

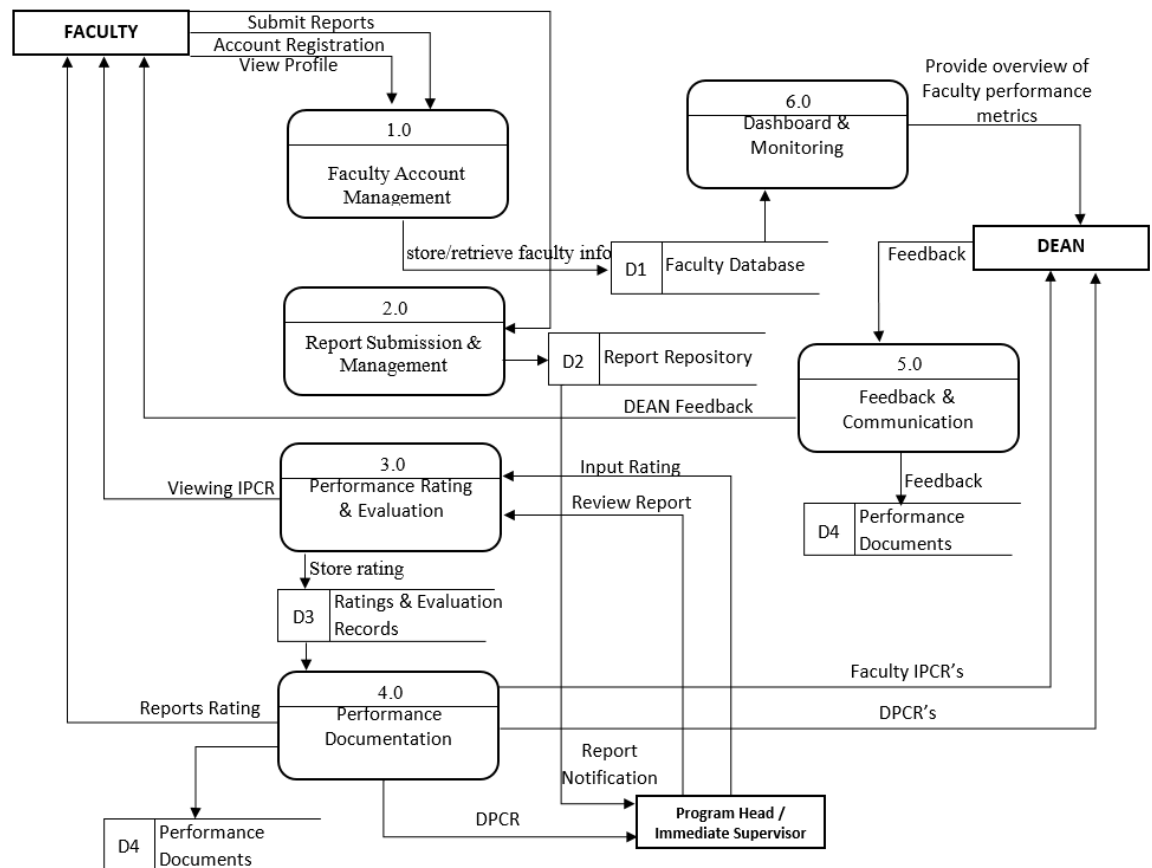


Figure 5. Data Flow Diagram (DFD – Level 1)

Entities

- Faculty – The primary users who submit reports, manage their profile and receive feedback.
- Dean - Oversees faculty performance, and provide feedback
- Program Head / Immediate Supervisor – Review submitted reports.

Process

1. **Faculty Account Management** - Handles registration, login, and profile management. It ensures secure access and maintains faculty credentials.
2. **Report Submission & Management** - Facilitates the uploading and tracking of performance reports. It routes submissions to evaluation and documentation modules.
3. **Performance Rating & Evaluation** - Stores evaluator scores and comments. This module supports rating computation and historical tracking of performance metrics.
4. **Performance Documentation**-Generates formal evaluation documents based on submitted reports and ratings. It also sends notifications to supervisors.
5. **Dashboard & Monitoring** - Provides real-time visual summaries of faculty performance, including trends, completion status, and evaluation outcomes.
6. **Feedback & Communication** Enables two-way communication between faculty, evaluators, and administrators. It collects feedback, reviews reports, and supports transparency.

Data Store

- **Faculty Database** - Central repository for faculty profiles, credentials, and historical performance records.
- **Evaluation Records Store** - Contains rating data, evaluator comments, and computed scores used for documentation and analysis.
- **Performance Document Archive** - Stores generated reports (e.g., IPCR, DPCR) and institutional summaries for compliance and audit purposes.

Data Flow

- **Faculty Data Flow** - Faculty information flows from Faculty Account Management to Report Submission & Management and Dashboard & Monitoring.
- **Report Submission Flow** Submitted reports are routed to Rating & Evaluation Records and Performance Documentation for scoring and document generation.
- **Evaluation Feedback Flow** Ratings and comments flow from evaluators to Feedback & Communication, which then informs both Faculty and the Dean.
- **Documentation Flow** Finalized performance documents are sent to Program Heads/Supervisors and archived for institutional use.

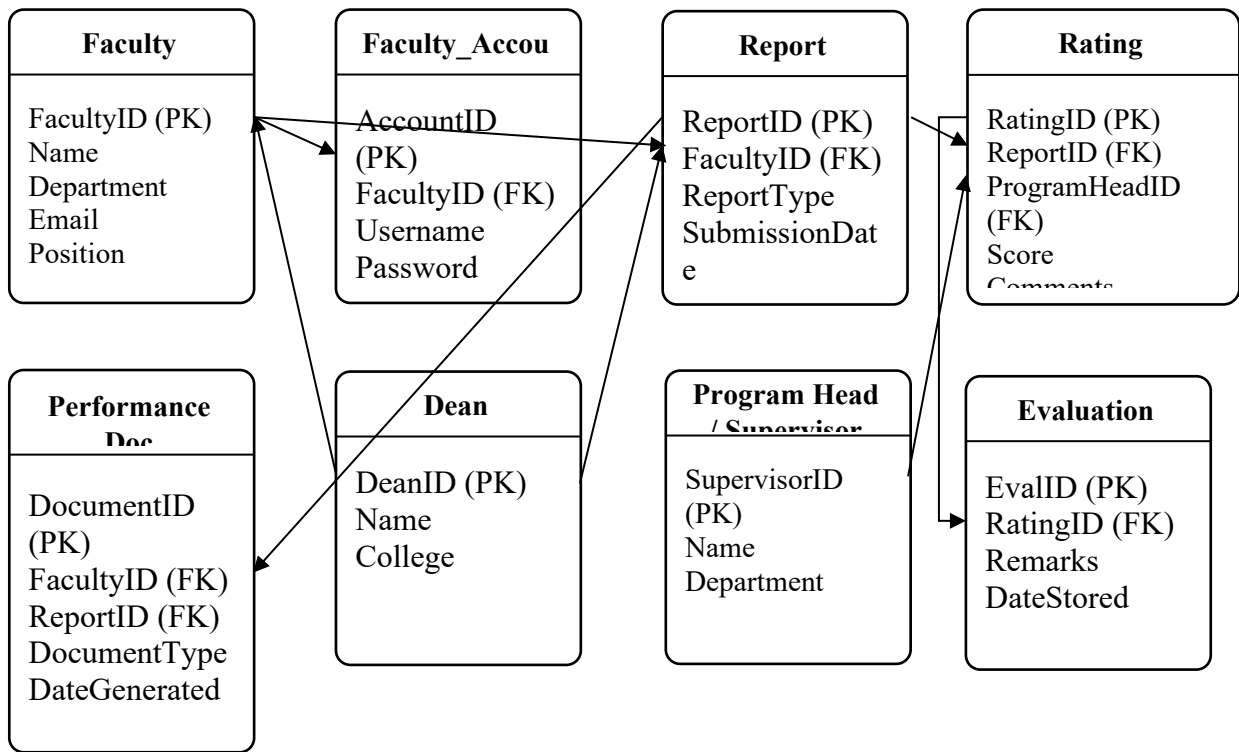


Figure 6. Entity – Relationship Diagram (ERD)

Faculty

- Faculty ID (PK)
- Name
- Department
- Email
- Position

This is the individual faculty members. Each faculty is identified by FacultyID as the Primary Key and associated with department and position. This entity will serve as the core reference for linking accounts, reports, and performance documentation.

Faculty Account

- AccountID (PK)
- FacultyID (FK)
- Username
- Password
- Role

This will capture every Performance reports submission. Each report is linked to a faculty member and includes metadata like type, submission data, and file location. This is a Many to one relationship with the faculty.

Rating

- ReportID (PK)
- FacultyID (FK)
- ReportType
- SubmissionDate
- FilePath

This entity contains the evaluation scores and comments provided by the program Head / Immediate supervisor.

Performance Documents

- DocumentID (PK)
- FacultyID (FK)
- ReportID (FK)
- DocumentType DateGenerated

This will represent the formal performance documentation generated from submitted reports and ratings. A many to one relationship is applied since a faculty can have multiple documents derived from different reports.

Dean

- DeanID (PK)
- Name
- College

This is the institutional authority overseeing the faculty performance. Though Deans are not directly linked to evaluation, they send and receive feedback and reports.

Program Head / Supervisor

- SupervisorID (PK)
- Name
- Department

They are the evaluators responsible for rating faculty reports. They provide scores and comments that feed into the rating and evaluation record entities. A one to many relationship is applied to this since a Program Head / Supervisor can evaluate many faculty.

Evaluation Record

- EvalID (PK)
- RatingID (FK)
- Remarks
- DateStored

Stores finalized evaluation records, including remarks and archival date. It links directly to the Rating entity and serves as a historical log for institutional review.

Development and Testing

System Development Model: This project uses **Iterative Waterfall Model**. This is a structured software development methodology that follows a sequential progression of phases—requirements, design, implementation, testing, deployment, and maintenance—

while allowing feedback and refinement between iterations. This model has been selected because it provides a balance between structure and flexibility, which is appropriate for the project's academic setting and the small number of the development team.

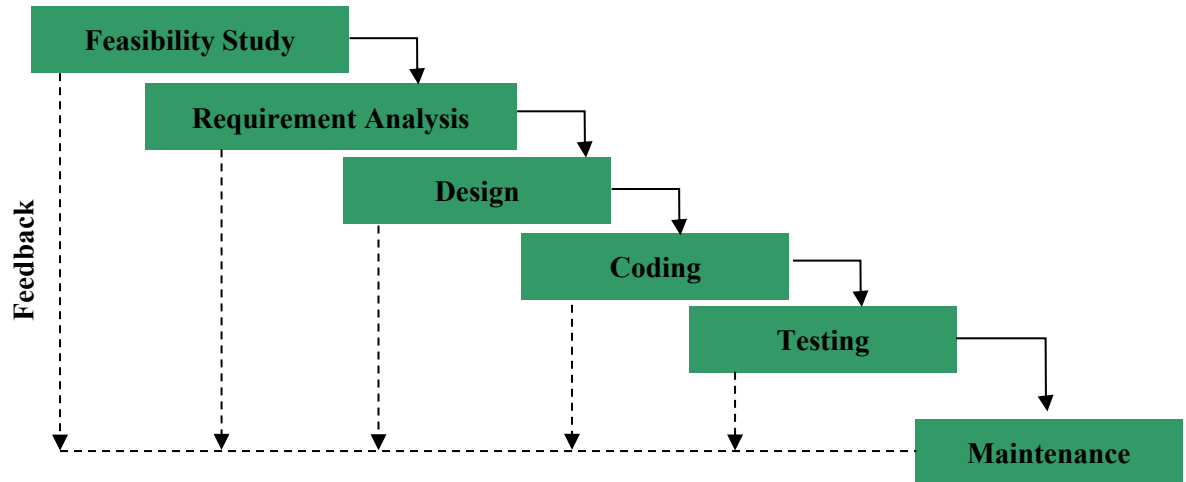


Figure 7. Iterative Waterfall Model

Phases/Iterations:

1. Requirements Analysis

Purpose: This phase gathers information and defines functional and non-functional requirements of the system.

Activities and Procedures: Stakeholder interviews were conducted to make sure that we are following CHED SPMS criteria, and reviewed the existing evaluation workflow.

Tools or Techniques Used: Stakeholder consultation forms and requirement specification documents.

Stakeholder Involvement: Faculty members, Dean, and institutional evaluators provide input on evaluation criteria, workflows, and reporting needs.

Deliverables or Outputs: Finalized requirement specification document.

2. System Design:

Purpose: To create users interface design based on the gathered requirements.

Activities and Procedures: Create diagrams and flowcharts to visualized system components and data relationships.

Tools and Techniques Used: MS Word, Lucidchart and Database schema Design.

Stakeholder Involvement: UI feedback was gathered from faculty to adjust layouts.

Deliverables or Outputs: System Diagram, UI mockups and database design documentation.

3. Development and Implementation:

Purpose: Building the system based on the approve design.

Activities and Procedures: Working with backend, frontend interface and integrated database operations.

Tools and Techniques Used: PHP, Javascript, MySQL, HTML, VS Code and Bootstrap.

Stakeholder Involvement: Giving update for milestone tracking.

Deliverables or Outputs: Functional system modules including login, report submission, ratings input, and report generation.

4. Testing and Refinement:

Purpose: To validate system functionalities, identify possible bugs, and ensure reliability before the deployment.

Activities and Procedures: Conduct unit testing, integration testing, and user's acceptability walkthroughs.

Tools and Techniques Used: Manual test scripts, Likert-scale survey for usability feedback.

Stakeholder Involvement: The faculty and Dean participated on the testing and provide feedback for improvement.

Deliverables or Outputs: Test case documentation, bug reports and updated system modules based on feedbacks.

5. Deployment:

Purpose: To launch the system in a live environment for college use.

Activities and Procedures: Configure cloud hosting, setting up domain access, and coordinated rollout with IT personnel.

Tools and Techniques Used: Cloud deployment services, SSL configuration, and institutional domain integration.

Stakeholder Involvement: Dean and IT staff facilitated deployment and initial user onboarding.

Deliverables or Outputs: Live system accessible via institutional domain, user accounts activated, and training materials distributed.

6. Documentation and Maintenance:

Purpose: To prepare formal documentation and ensure system sustainability post-deployment.

Activities and Procedures: Finalized user manuals, technical documentation, and disaster recovery protocols.

Tools and Techniques Used: Microsoft Word, and PDF export tools.

Stakeholder Involvement: Documentation lead coordinated with the project manager and QA tester to ensure completeness.

Deliverables or Outputs: Capstone manuscript, user's manual, training guides, and maintenance plan.

Software Development Tools

The Faculty Performance Documentation and Evaluation System was developed using a suite of modern and reliable open source technologies to ensure scalability, maintainability, and user centered design. Below are the details:

- **HTML:** Used for structuring web content and ensuring semantic clarity across pages.

- **PHP:** Serves as the server-side scripting language, handling backend logic, database interactions, and dynamic content rendering.
- **JavaScript (jQuery/Ajax):** Enhances client-side interactivity, supports asynchronous data exchange, and streamlines DOM manipulation for responsive user experiences.
- **MySQL:** A robust relational database system used to store, retrieve, and manage structured data related to faculty profiles, performance metrics, evaluation records, and system logs. It supports ACID compliance and integrates seamlessly with PHP.
- **Visual Studio Code (VS Code):** A lightweight yet powerful code editor offering syntax highlighting, version control integration (Git), and extensions for PHP, JavaScript, and MySQL development. It facilitated efficient coding, debugging, and collaboration.
- **Bootstrap:** A front-end framework used to design responsive and mobile-first interfaces. It provided pre-built UI components, grid systems, and styling utilities that accelerated development while ensuring visual consistency and accessibility.

Test cases and Test Plan

Test Plan Overview: The test plan summarizes the organized method used to validate functionality, reliability, and performance of the Faculty Performance Documentation and Evaluation System. It defines the scope of testing, identifies components for evaluation, specifies testing types, and assigns responsibilities to ensure that the system meets institutional standards and users' expectations prior to deployment.

Components: The system contains several modules such as Faculty Account Management, Report Submission, Rating and Evaluation Records, Performance Documentation, Dashboard and Monitoring, and Feedback and Communication. Every component will go through testing to check how they behave.

Scope of Testing: All essential modules and user interfaces will be tested. Workflows for login and registration, report submission and retrieval, rating computation logic, document production, dashboard analytics, and feedback mechanisms are all included in this. With regards to data security and integrity, role-based access and administrative restrictions will also be observed.

Objectives: The primary goals of testing are to ensure that all system functionalities operate as intended, identify and resolve bugs or inconsistencies, validate performance under concurrent usage, and confirm compliance with institutional data handling protocols. Testing also aims to verify that the system is user-friendly, secure, and scalable.

Testing Types:

- **Unit Testing:** Validate individual functions and modules (e.g., login, rating calculator).
- **Integration Testing:** Assure smooth interaction between modules.
- **System Testing:** Examine the system as a whole in a production-like setting.
- **Performance Testing:** Assess system responsiveness, load handling, and data processing speed.
- **Security Testing:** Verify encryption protocols, access control, and protection against unauthorized access.

Test Cases

Test Case ID	Feature/Module	Test Description	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC-01	Faculty Login	Verify login with valid credentials	Faculty account exists and active	1.Go to log in page 2.Enter valid username /password 3.Click log in	Redirect to Dashboard		
TC-02	Invalid Login	System verification of incorrect credentials	Faculty account exist	1.Enter wrong password 2.Click log in	"Invalid credentials" message displayed		
TC-03	Password Reset	Verify that faculty can reset their password via email	Valid email registered in system	1.Click "Forgot Password" 2.Enter registered email 3.Check inbox and follow reset link	Password reset successfully		
TC-04	Account Lockout After Failed Attempts	Ensure system locks account after multiple failed login attempts	Faculty account exists	Enter wrong password 5 times	Account temporarily locked		
TC	Account Unblocked by the Admin						
TC-05	Role-Based Access Control	Verify that faculty cannot access admin-only pages	Logged in as faculty	Attempt to access admin dashboard URL	Access denied or redirected		
TC-06	Session Timeout	Validate automatic logout	Logged in	Leave session idle for	User is logged out	Ask how man	

		after inactivity		configured timeout duration		y min/ hours	
TC-07	Profile Update	Faculty updates personal information	Logged in	1.Navigate to profile 2.Edit name or email 3.Save Changes	Profile updated successfully		
TC-08	Admin Account Creation	Admin creates a new faculty account	Logged in as admin	1.Go to “Manage Users” 2.Enter faculty details 3.Assign role 4.Save	New account created and listed		
TC-09	Login Audit Trail	Verify that login attempts are logged	System logging enabled	1.Login as faculty 2.Check audit log	Timestamped entry recorded		
TC-10	IPCR Report Upload	Faculty uploads document	Logged in	1.Go to “Submit Report” 2.Upload file 3.Click “Submit”	Confirmation message		
TC-11	Duplicate Submission	Prevent duplicate IPCR uploads	Report already submitted	1.Attempt to upload same file again.	Warning or overwrite prompt		
TC-12	Rating Computation	Validate correct rating calculation based on inputs	IPCR data submitted	1.Admin initiates evaluation 2.System computes rating	Accurate rating displayed		
TC-13	Rating Override	Admin manually adjusts rating	Rating computed	1.Admin edit score 2.Save changes	Updated rating saved		
TC-14	Export to PDF	Generate PDF of submitted report	Report submitted	1.Click Export 2.Select Format	Downloadable PDF generated		
TC-15	Print Preview	Display printable version of report	Report submitted	1.Click “Print Preview”	Formatted view shown		

TC-16	Add Evaluation Period	Admin creates new evaluation cycle	Logged in as admin	1.Go to “Manage Periods” 2.Enter dates 3.Save	New period added		
TC-17	User Role Assignment	Assign faculty/admin roles	User account exists	1.Select User 2.Assign role	Role updated		
C-18	Suggested Decision Review	Verify that means can view suggested ratings and comments from Program heads	Program Head has submitted evaluation; mean is logged	.Navigate to “Suggested Decisions” dashboard .Select evaluation period and faculty name .Review suggested rating and remarks	Suggested decision is displayed with full context (rating, justification, attachments)		
C-19	Post-Decision Finalization	Verify system behavior after mean submits final decision	mean has submitted final decision	.System updates evaluation status to “Finalized” .Locks evaluation record from further edits .Sends notification to faculty .Archives decision in audit trail	evaluation status changes to “finalized”; record becomes read-only; faculty notified; audit trail updated		

Table 1. System Test Cases

User Acceptance Testing (UAT):

UAT Objectives

- Validate that the system supports core business processes related to faculty performance documentation, evaluation, and reporting.
- Ensure user satisfaction across faculty and administrative roles.
- Confirm system readiness for deployment in alignment with institutional.
- Identify usability issues and gather feedback for final refinements.

Participants

- **College of Arts and Sciences Faculty Members** – Submit IPCR/DPCR reports, view evaluation results
- **Department Chairs/Program Heads** – Review submissions, assign ratings
- **Dean, College of Arts and Sciences** - Oversee compliance and deployment readiness

UAT Scenarios/Scripts

ID	Task Description	User Role
UAT-001	Submit report with attachments	Faculty
UAT-002	View submission status and history	Faculty
UAT-003	Review and rate submitted reports	Department Chair
UAT-004	Generate summary reports for evaluation period	Dean
UAT-005	Test access control and role-based permissions	Admin/Dean
UAT-006	Simulate evaluation cycle from submission to final rating	Admin/Dean

Acceptance Criteria

- All critical workflows (submission, review, rating, reporting) function without errors
- Role-based access and permissions are correctly enforced
- Submitted documents are securely stored and retrievable
- Evaluation results are accurately computed and displayed
- System performance is stable under expected load
- **Users report $\geq 90\%$ satisfaction in usability and functionality surveys**
- No critical bugs remain unresolved at the end of UAT

UAT Schedule

Phase	Dates	Duration
Preparation & Script Finalization	Oct 6–10, 2025	5 days
UAT Execution	Oct 13–17, 2025	5 days
Feedback Consolidation & Fixes	Oct 20–23, 2025	4 days
Final Sign-off	Oct 24, 2025	1 day

Deployment Plan

The system will be deployed using a phased rollout strategy which will begin with a pilot implementation in selected departments before the College wide deployment. This approach allows for controlled testing, feedback collection, and incremental adjustments, minimizing disruption and ensuring user readiness. The pilot phase will validate core functionalities such as report submission, rating workflows, and dashboard monitoring before expanding access to all faculty and administrative users.

Installation Procedure: The following are the steps for system installation:

1. Server Preparation

- Provision a secure cloud-based hosting environment
- Configure PHP, MySQL, and Apache/Nginx stack
- Enable SSL/TLS encryption for secure data transmission
- 2. System Deployment
 - Upload source code via FTP
 - Set up database schema using MySQL scripts
 - Configure environment variables and access credentials
 - Test connectivity and module responsiveness
- 3. User Access Setup
 - Create initial admin and faculty accounts
 - Assign roles and permissions based on institutional hierarchy
 - Enable login portal and dashboard access
- 4. Final Validation
 - Conduct smoke testing to verify core functions
 - Confirm report submission, rating, and documentation modules are operational

Data Migration Plan: The system migration will focus more on digitalizing the manual and spreadsheets process of faculty records and performance evaluation.

- Data Collection
 - Gather historical IPCR/DPCR files and evaluation records from Department
 - Use standards format for database compatibility
- Data Cleansing
 - No duplicate data, inconsistency will be corrected, and validated entries
 - Ensure alignment with faculty IDs and department codes.
- Migration Execution
 - Use structured import scripts to populate MySQL tables
 - Verify integrity through sample queries and dashboard previews
- Post Migration Validation
 - Conduct spot checks with department Head
 - Confirm completeness and accuracy of migrated data

Cutover Strategy: The college has no previous system

Rollback Plan: If ever there will be an unresolved issues during deployment, the following rollback plan will be executed:

- Immediate Suspension: Temporary suspension system access and revert to manual submission, and notify all stakeholders via email
- Data Restoration: Restore the backup records and spreadsheets, re-enable the previous documentation workflow.
- Issue Resolution: Diagnose the root cause of the issue, apply repairs and retest the module.
- Post-Rollback Review: Document the incident for future use, and preventive measure. Update deployment checklist and contingency protocols.

Risk Management Plan

Risk Id	Risk Description	Likelihood	Impact	Mitigation Strategy	Contingency Plan	Responsible Person
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R-01	Scope creep due to evolving stakeholder demands	Medium	High	Clearly define the project scope and implement change control procedure.	Review the schedule and available resources then give priority to the important elements	Project Manager
R-02	Technical Challenges during integration of Module	Medium	Medium	Perform modular testing, use tested framework	Separate faulty module; rollback to stable version	Developer
R-03	Budget overruns due to unforeseen infrastructure needs	Low	High	Maintain budget buffer, and monitor weekly expenses	Seek college support	Project Manager
R-04	Low User adoption due to unfamiliarities	High	High	Conduct Training workshops	Offer one-on-one testing	Training Coordinator
R-05	Data security breach during deployment or usage	Low	High	Implement SSL encryption and role-based access	Disable system in a moment and restore back up.	Developer
R-06	Delays in documentation and report preparation	Medium	Medium	Early documentation assignments and use templates.	Extend Documentation window	System analyst
R-07	Staff turnover affecting continuity	Low	Medium	Cross-train team members	Reassign roles	Project Manager
R-08	Server downtime during live deployment	Low	High	Test load capacity beforehand	Notify user of the temporary outage	Developer

Table 2. Risk Management Plan

Disaster Recovery Plan

Purpose: In the case of a significant disruption, such as a natural disaster, cyberattack, or hardware failure, to guarantee the quick restoration of system operation and data integrity. Both teacher performance records and institutional activities are safeguarded by this arrangement.

Key Personnel:

Project Manager – Overall coordinator and decision maker.

Developer- Technical recovery and system restoration

IT Support Staff – Server access, backup retrieval, and network restoration

Deployment Coordinator – Communication with stakeholder and rollback execution

Backup Strategy:

Frequency: Daily incremental backups; weekly full backups

Location: Cloud Storage

Type: Encrypted MySQL database dumps and source code archives

Retention: 30-day rolling backup cycle with offsite replication |

Recovery Procedure:

1. Assess the nature and scope of the disruption
2. Inform key personnel and initiate recovery protocol
3. Restore latest backup from cloud
4. Reconfigure server environment and redeploy application
5. Test system modules and verify data integrity
6. Resume operations and notify users of restoration status

Recovery Time Objectives: Maximum acceptable downtime is 48 hours from the start of disruption.

Recovery Point Objectives (RPO): Maximum acceptable data loss is 24 hours based on daily backup frequency.

Testing Schedule:

- Quarterly DRP Simulation: Perform mock recovery drills quarterly
- Annual Full Recovery Test: end-to-end restoration procedures must be checked.

Users Training Plan

Training Objectives: The training program is for the end-user to have the confidence to use the system. Upon the end of this training, the user must be able to:

- Able to log in securely and navigate the system interface.
- Able to submit required document report
- Monitor submission status and evaluation result
- Perform role-specific tasks such as rating, reviewing, and generating reports
- Able to do basic troubleshooting and access help resources

Target Audience:

Faculty members – for report submission and status tracking

Program head/Supervisor – for reviewing and rating of documents

Deans – for oversight, summary reporting, and policy alignment

Training Content:

- System overview and purpose
- User interface (dashboard, forms, and notifications)
- Step by step guide in submitting reports
- Rating and evaluation workflows
- Exporting and Importing documents
- Data privacy and access controls
- Troubleshooting and support channels

Training Methods

- Departmental Workshops – Hands on sessions with live system access
- User Manuals – printed and digital guides
- One-on-One sessions – for personal support
- Video Tutorials – for asynchronous learning
- Helpdesk support – for day to day assistance

Training Schedule

Date	Audience	Format	Location
Oct 21, 2025	Faculty Members	Workshop	ICT Lab
Oct 22, 2025	Program Heads/Supervisors	Workshop	CAS Bldg. Rm 2
Oct 23, 2025	Deans	Briefing & Demo	Dean's Office
Oct 24–29, 2025	All Users	One-on-One & Helpdesk	On-call & Online

Table 3. Training Schedule

Training Materials

- Printed user manuals and quick-start guides
- Slide presentations (system overview, workflows)
- Practice exercises (sample submissions, rating simulations)
- FAQ sheets and troubleshooting checklists

Evaluation Method

Training effectiveness will be measured through:

- Post-training feedback forms
- Task-based assessments
- Usage monitoring during pilot phase
- Follow-up interviews with department heads to assess confidence and adoption
- Helpdesk ticket analysis to identify recurring issues

User's Manual

Faculty Performance Documentation and Evaluation System

Table of Contents:

Introduction

Getting Started

Core Functionalities

Dashboard Overview

IPCR submission

Evaluation Rating

Report Generation

Troubleshooting

Frequently asked questions (FAQ)

Glossary

Introduction

This document gives the users with the detailed guide to use the Faculty Performance Documentation and Evaluation System. Faculty performance document submission, evaluation, and reporting are made easier by the system architecture which guarantee timeliness, transparency and adherence to institutional standards.

Getting Started

Access instruction:

- Open a web browser and type in the official system URL provided.
- Recommended browsers: Chrome, Firefox, Edge.

Login Procedure:

- Enter your assigned username and password.
- First-time users will be prompted to change their password and accept the data privacy agreement.
- If login fails, use the “Forgot Password” link or contact your department’s system coordinator.

Core Functionalities

Dashboard Overview

- It displays the submission status, evaluation results, and the system notifications.
- Navigation menu includes: Home, Submit Report, View Ratings, and Export Documents.

Report Submission

Steps:

1. Click “Submit Report” from the dashboard.
2. Select the evaluation period and document type.
3. Upload your completed form in PDF format.
4. Add supporting documents if required.
5. Click “Submit” and wait for confirmation.

Evaluation and Rating (For Program Heads/Supervisors)

Steps:

1. Navigate to “Review Submissions.”
2. Select faculty name and evaluation period.
3. View submitted documents.
4. Enter ratings based on approved metrics.
5. Save and finalize evaluation.

Report Generation

Steps:

1. Go to “Export Documents.”
2. Choose report type (e.g., Summary, Individual Rating).
3. Select format (PDF or Excel).
4. Click “Generate” and download

Troubleshooting

Issue	Solution
Cannot log in	Check credentials, use “Forgot Password” and or contact admin
File upload fails	Make sure that the file is in PDF format and under size limit

Issue	Solution
Ratings not saving	Confirm all required fields are filled; refresh page
Dashboard not loading	Clear browser cache; try a different browser

Frequently Asked Questions (FAQ)

Q1: Can I edit a submitted report?

A: Yes, if the evaluation period is still open. Go to “My Submissions” and click “Edit.”

Q2: Who can see my evaluation results?

A: Only authorized personnel such as your supervisor and dean.

Q3: What if I miss the submission deadline?

A: Contact your program head immediately. Late submissions may be flagged.

Q4: Is my data secure?

A: Yes. The system uses encrypted connections and role-based access controls.

Glossary

IPCR: Individual Performance Commitment and Review

DPCR: Department Performance Commitment and Review

SPMS: Strategic Performance Management System

Dashboard: Main interface showing user-specific data

Evaluation Period: Designated timeframe for submitting and reviewing reports

Rating: Quantitative assessment of performance based on institutional metrics